

Empowering Compute, Network, Storage & AI

AI Server Hardware Validation Report

8x NVIDIA B200 - GPU, Compute, Memory, NVMe, Network, Power & Thermal

Server / Host	ty-cam-38 (172.16.15.38)
Platform	Tyrone Camarero AI / 2x Intel Xeon 6788P + 8x NVIDIA B200 (HGX)
Validation Tier	Qualification
Campaign Window	2026-06-01 -> 2026-06-04
Prepared For	Netweb Technologies India Limited
Date	2026-06-05

Overall Result: PASS WITH LIMITATIONS

Executive Summary

This report covers a tier-2 (qualification) validation of a Tyrone Camarero AI node: dual Intel Xeon 6788P, 2 TB DDR5, eight NVIDIA B200 (HGX) accelerators on an NVLink fabric, four Samsung PM9D3a NVMe drives, and 400 GbE/NDR networking. The GPU, compute, memory, network and power/thermal subsystems meet or exceed datasheet expectations under sustained combined load.

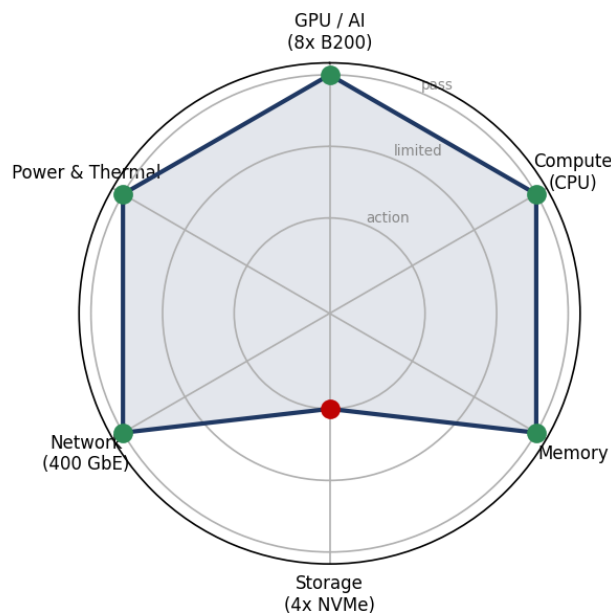
All eight B200 GPUs sustained ~4.41 PFLOPS FP8 each (98% of datasheet) with full NVLink bandwidth, zero ECC errors, and DCGM level-3 diagnostics passing. Peak wall power reached 12.6 kW at a 25 C inlet with no GPU thermal throttling.

One storage finding requires action: drive nvme3 negotiated a PCIe Gen5 x2 link instead of the capable x4, dropping its 4K random write to 53% of rating. The other three drives are within spec; re-seat / bifurcation remediation and a re-test are recommended.

Validation Scorecard

Subsystem	Result	Detail	Verdict
GPU / AI (8x B200)	GEMM FP8 ~4.41 PFLOPS/GPU	98% of datasheet; NVLink 1.76 TB/s; 0 ECC; DCGM L3 pass	PASS
Compute (CPU)	HPL 6.10 TFLOPS	90% of Rpeak; stable across soak, 0 throttle events	PASS
Memory	STREAM Triad 920 GB/s	DDR5-6400, all DIMMs at rated speed; loaded latency 118 ns	PASS
Storage (4x NVMe)	3 of 4 drives at spec	nvme3 negotiated PCIe x2 -> random write 53% of rating	ACTION
Network (400 GbE)	RoCE 392 Gb/s	Near line-rate RoCEv2; PFC + DCQCN verified lossless	PASS
Power & Thermal	Peak 12.6 kW @ 25 C inlet	Outlet 52 C, fans 19.5k RPM at combined load; no throttle	PASS

Subsystem health at a glance



Hardware Inventory

System

Vendor	Tyrone (Netweb)
Model	Camarero AI HGX-B200
Board	auto-detected at run time (dmidecode)
Chassis	8U HGX
Hostname	ty-cam-38
Serial	TY-CAM-380417
CPU	
Model	2x Intel Xeon 6788P (Granite Rapids)
Sockets	2
Cores Per Socket	86
Threads	344
Base Ghz	2.0
Boost Ghz	3.8
Numa Nodes	2
Isa	AVX-512, AMX, BF16
Numa Device Map	GPU0-3 -> node0; GPU4-7 -> node1; nvme0-3 -> node0
Memory	
Total	2 TB
Type	DDR5 RDIMM
Dimms Populated	32 x 64 GB (16 per socket)
Configured Speed	6400 MT/s
Rated Speed	6400 MT/s
Speed Status	all DIMMs at rated speed
BIOS / Platform	
Version	2.4
Date	2026-02-10
Power Profile	Max Performance
C States	disabled
Numa Nps	NPS2
Pcie Aspm	disabled
M2 Bifurcation	x4x4x4x4 (slot 6 reports x2 -- see findings)
BMC / Power	
Firmware	BMC 2.06.0
Psu Model	16 kW shelf (6x 3kW, 5+1)
Psu Watts	18000 W (5+1)
Redundancy	N+1 redundant
Telemetry Source	Redfish (DCMI power fallback)
Kernel / OS	
Distro	Ubuntu 24.04.1 LTS
Kernel	6.8.0-45-generic
Governor	performance
Tuned Profile	accelerator-performance

Cmdline ro quiet pcie_aspm=off nvme_core.default_ps_max_latency_us=0 iommu=pt nvidia_drm.modeset=1

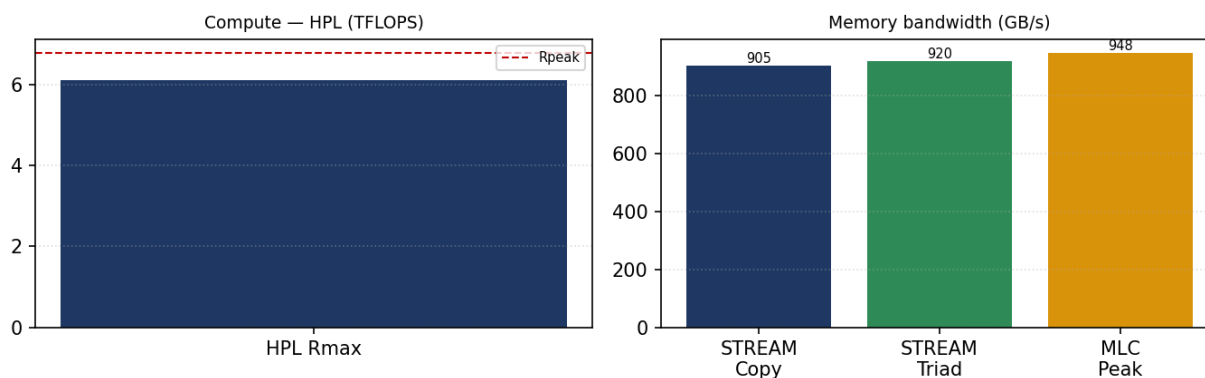
Storage Devices

Dev	Model	Form	Capacity	Serial	Link (neg)	Link (cap)
/dev/nvme0n1	Samsung PM9D3a	E3.S	3.84 TB	S6X1A0	Gen5 x4	Gen5 x4
/dev/nvme1n1	Samsung PM9D3a	E3.S	3.84 TB	S6X1A1	Gen5 x4	Gen5 x4
/dev/nvme2n1	Samsung PM9D3a	E3.S	3.84 TB	S6X1A2	Gen5 x4	Gen5 x4
/dev/nvme3n1	Samsung PM9D3a	E3.S	3.84 TB	S6X1A3	Gen5 x2	Gen5 x4

Network Interfaces

Interface	Model	Speed	Driver
ibp1s0	NVIDIA ConnectX-7 NDR (400G)	400 Gb/s	mlx5_core 24.07
eno1	Intel X710	10 Gb/s	i40e

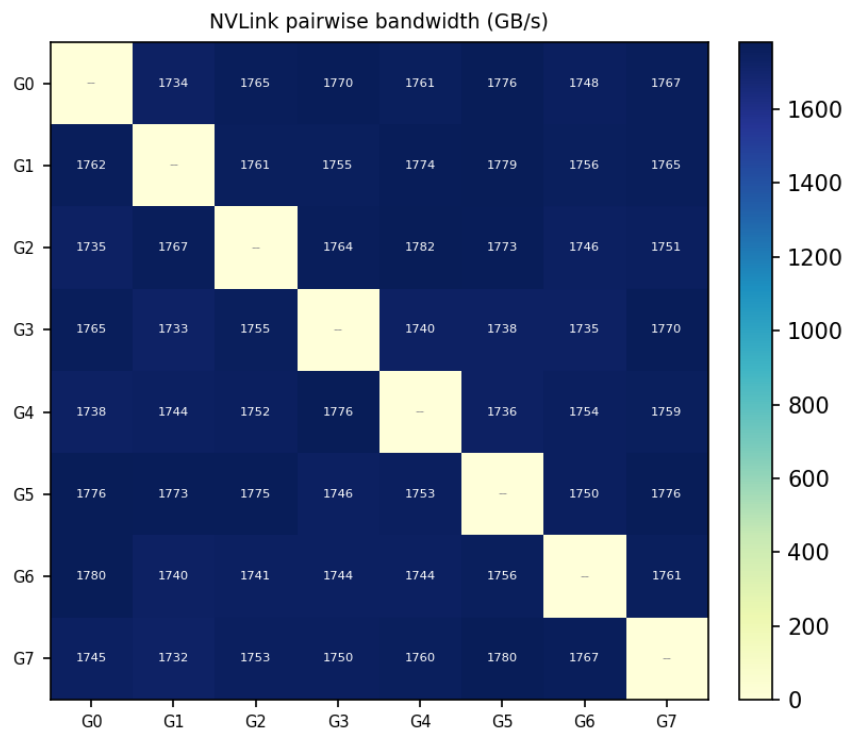
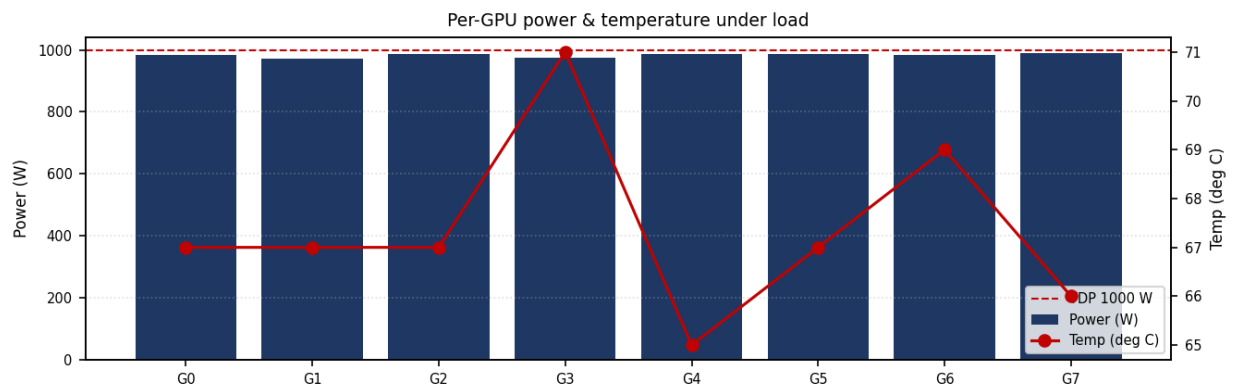
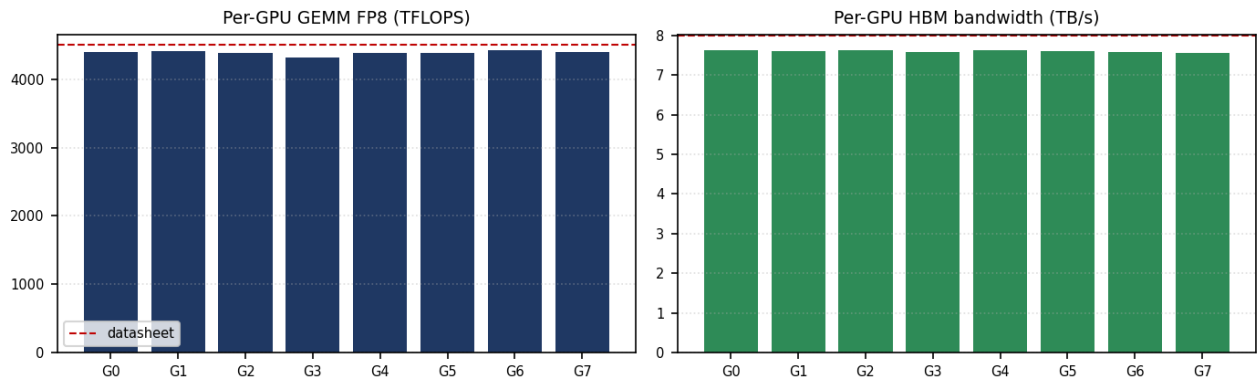
Compute & Memory

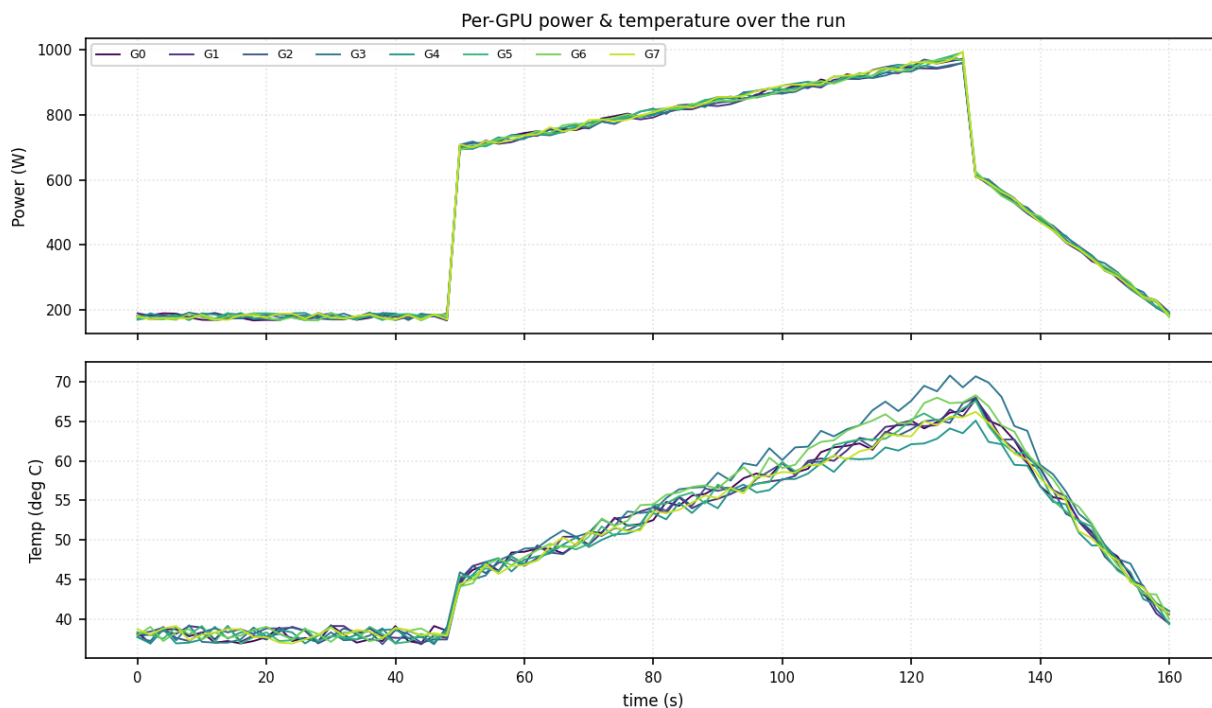


HPL Rmax (TFLOPS)	6.1
HPL efficiency (%)	90
AVX-512 / AMX	yes / yes
Soak (h) / throttle events	24 / 0
STREAM Triad (GB/s)	920
STREAM Copy (GB/s)	905
MLC peak BW (GB/s)	948
MLC loaded latency (ns)	118
DIMM speed (MT/s)	6400 (rated 6400)

GPU / AI Accelerators

8 × NVIDIA B200 (HGX) -- per-GPU GEMM throughput, HBM bandwidth, NVLink fabric, and power/thermals under sustained AI load.



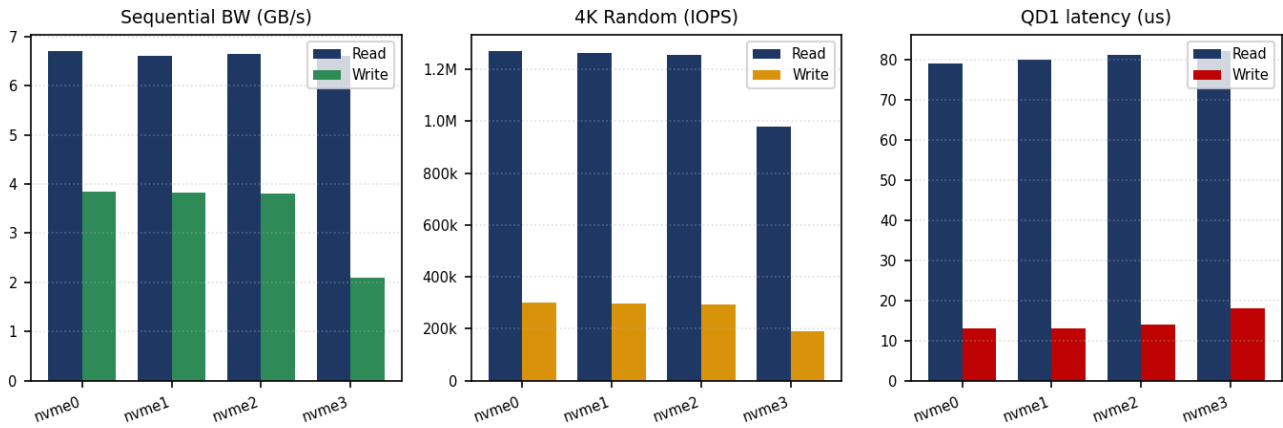


GPU	GEMM FP8	GEMM BF16	HBM BW	NVLink	Power	Temp	ECC
G0	4398	2226	7.6 TB/s	1.75 TB/s	985 W	67 C	0
G1	4411	2221	7.6 TB/s	1.75 TB/s	973 W	67 C	0
G2	4384	2229	7.6 TB/s	1.78 TB/s	987 W	67 C	0
G3	4318	2254	7.6 TB/s	1.75 TB/s	974 W	71 C	0
G4	4388	2243	7.6 TB/s	1.76 TB/s	986 W	65 C	0
G5	4389	2247	7.6 TB/s	1.75 TB/s	987 W	67 C	0
G6	4431	2248	7.6 TB/s	1.76 TB/s	985 W	69 C	0
G7	4395	2259	7.6 TB/s	1.76 TB/s	991 W	66 C	0

Units: GEMM in TFLOPS; HBM/NVLink bandwidth measured per GPU.

- NCCL all-reduce (8-GPU): 478 GB/s bus bandwidth.
- Aggregate HBM: 1.5 TB; driver 560.35.03, CUDA 12.6.
- DCGM diagnostics: passed (level 3, all subtests).

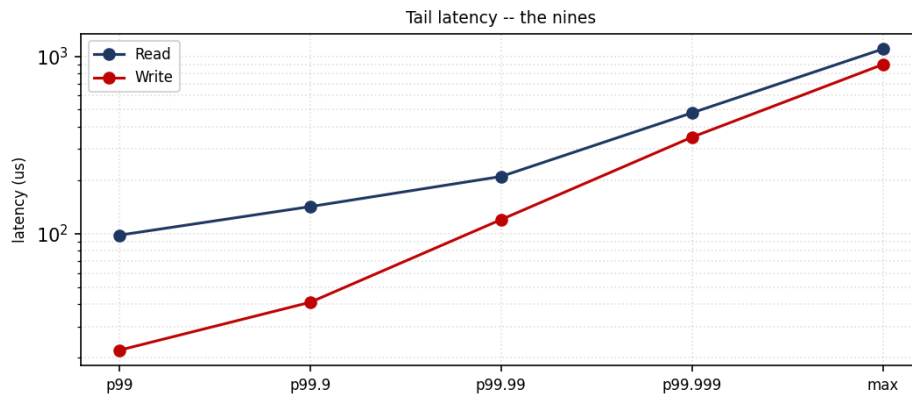
Storage Performance



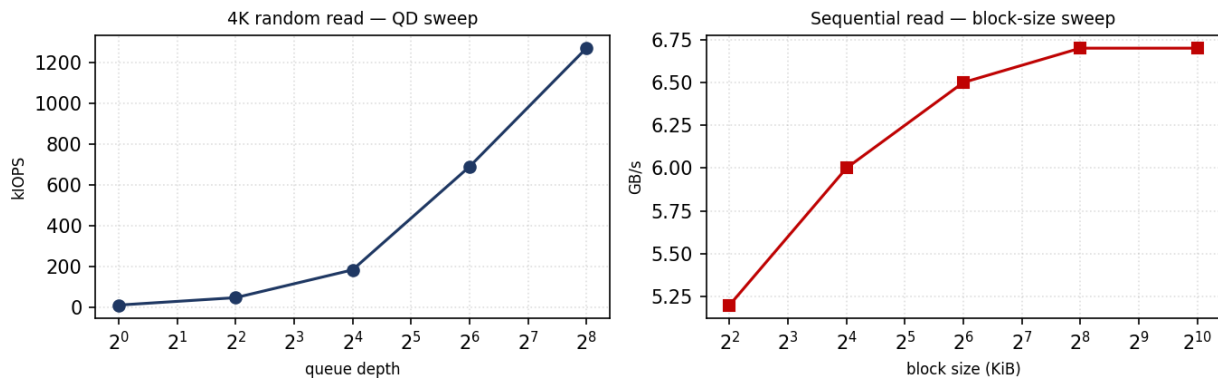
Drive	Link	SeqR	SeqW	RandR	RandW	QD1R	QD1W	p99.9R
nvme0	Gen5 x4	6.7	3.9	1.27M	300k	79	13	142
nvme1	Gen5 x4	6.6	3.8	1.26M	298k	80	13	148
nvme2	Gen5 x4	6.7	3.8	1.25M	295k	81	14	151
nvme3	Gen5 x2	6.6	2.1	980k	190k	82	18	233

Units: SeqR/SeqW in GB/s; RandR/RandW in IOPS; QD1 & p99.9 in microseconds.

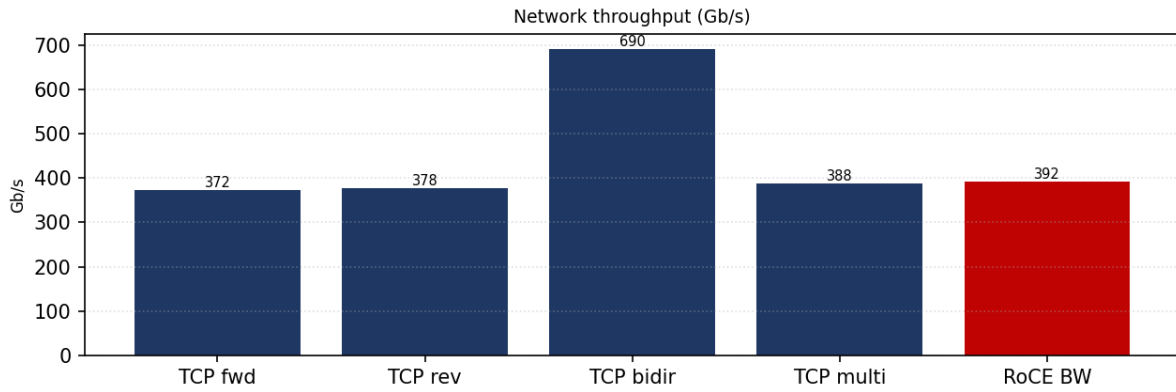
Tail Latency (representative drive)



Queue-depth & block-size sweeps



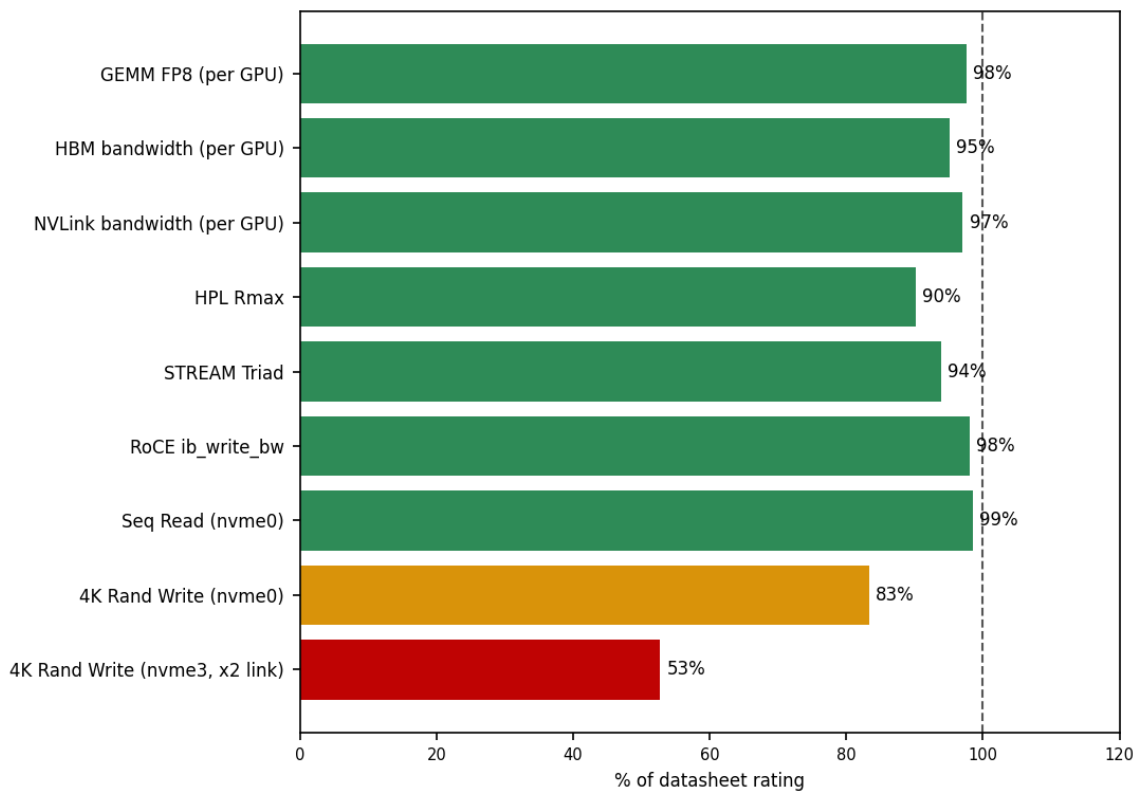
Network



TCP RX / TX (Gb/s)	372 / 378
TCP bidirectional (Gb/s)	690
RoCE ib_write_bw (Gb/s)	392
RoCE p99.9 latency (us)	3.4
MTU / PFC / ECN	9000 / enabled (prio 3) / DCQCN enabled
RDMA link	400 Gb/s, Active
Peer / fabric	ty-cam-37 / NDR via lossless switch

Specification Compliance

Measured performance as a percentage of datasheet rating. Reference line at 100%. Bars auto-colored: $\geq 90\%$ green, 75-90% amber, $< 75\%$ red.

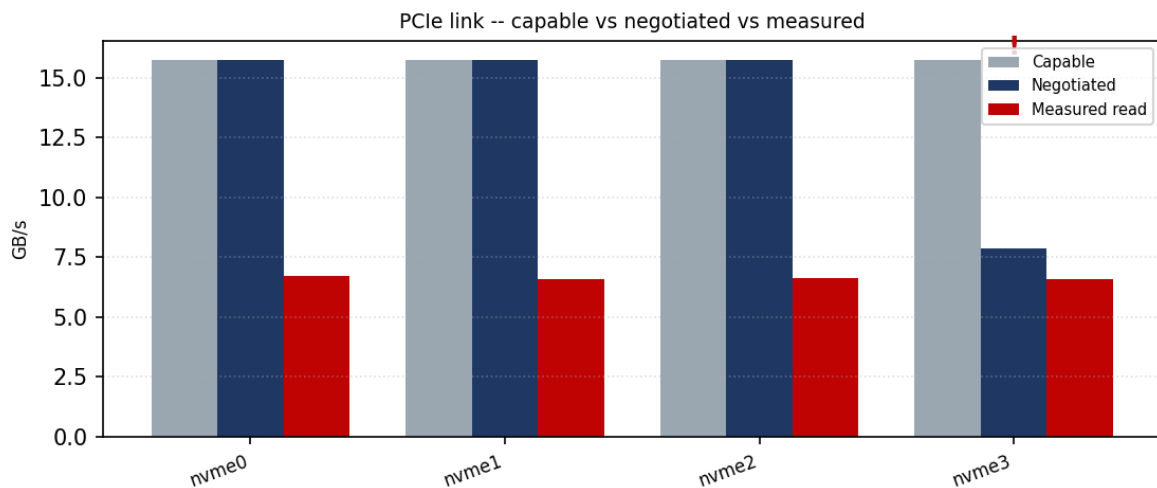


Metric	Measured	Rated	Unit	% of rating
GEMM FP8 (per GPU)	4,389	4,500	TFLOPS	98%

Metric	Measured	Rated	Unit	% of rating
HBM bandwidth (per GPU)	7,610	8,000	GB/s	95%
NVLink bandwidth (per GPU)	1,747	1,800	GB/s	97%
HPL Rmax	6.1	6.77	TFLOPS	90%
STREAM Triad	920	980	GB/s	94%
RoCE ib_write_bw	392	400	Gb/s	98%
Seq Read (nvme0)	6.7	6.8	GB/s	99%
4K Rand Write (nvme0)	300,000	360,000	IOPS	83%
4K Rand Write (nvme3, x2 link)	190,000	360,000	IOPS	53%

PCIe Link Analysis

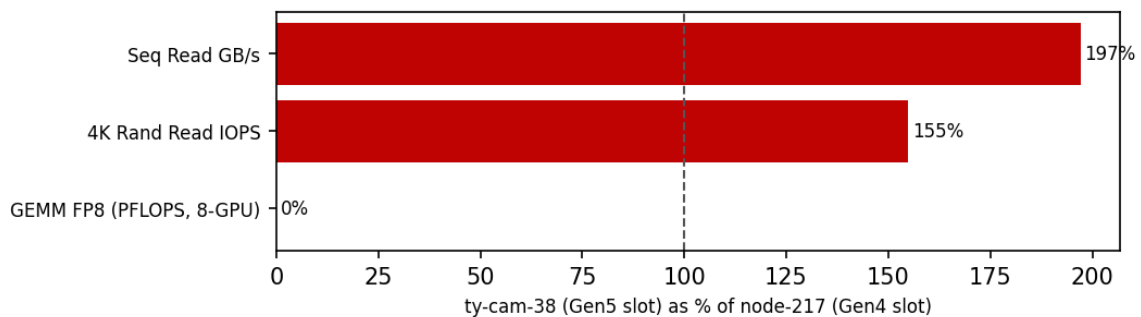
Capable vs negotiated link bandwidth vs measured sequential read. Drives negotiating below their capable width/speed are flagged.



Link warnings: nvme3: negotiated PCIe x2, capable x4 -- random write 53% of rating

Cross-Server Comparison

Same four PM9D3a drives across slot generations; the AI node adds 8x B200.



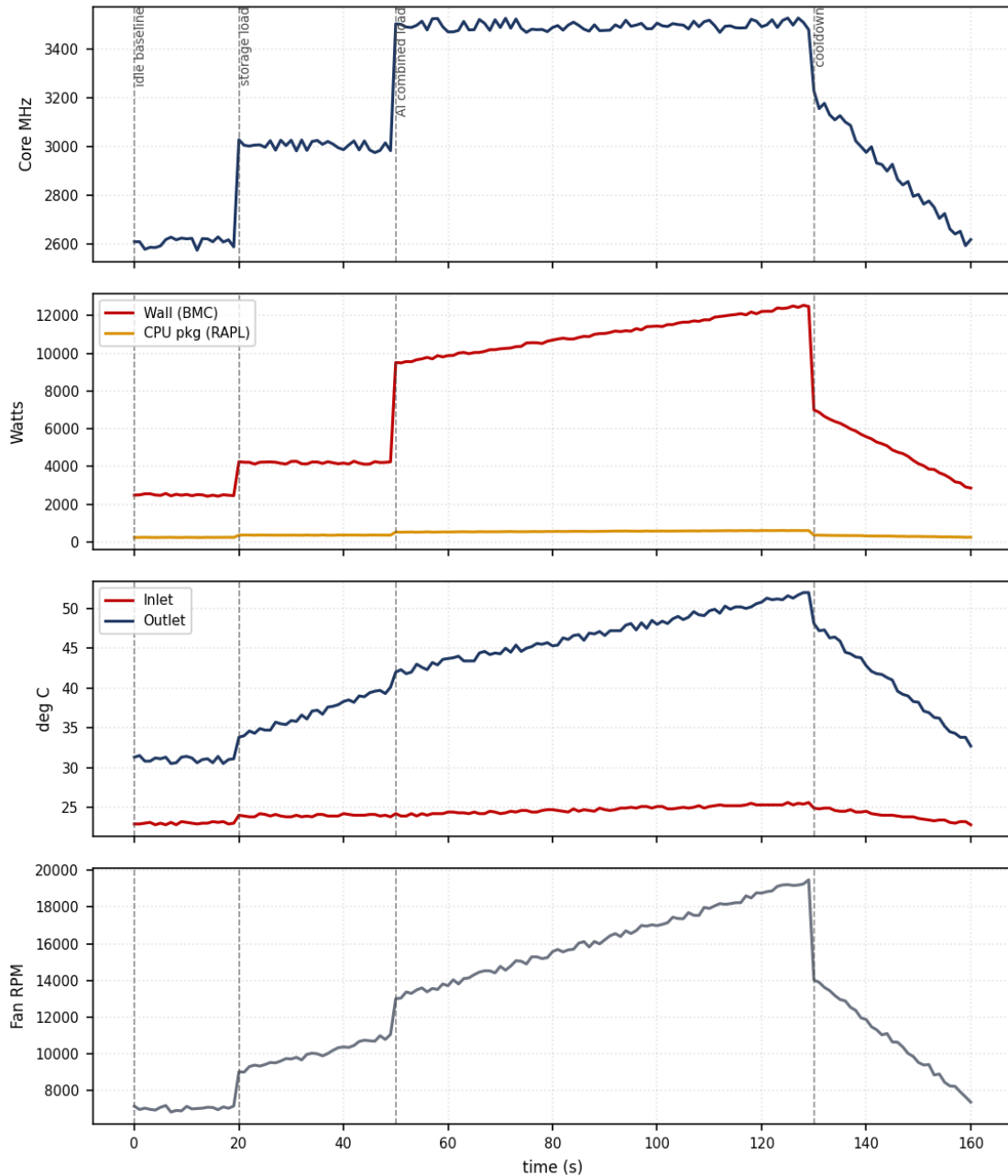
Metric	node-217 (Gen4 slot)	ty-cam-38 (Gen5 slot)	Delta
Seq Read GB/s	3.4	6.7	+97%
4K Rand Read IOPS	820,000	1,270,000	+55%

Metric	node-217 (Gen4 slot)	ty-cam-38 (Gen5 slot)	Delta
GEMM FP8 (PFLOPS, 8-GPU)	0.0	35.3	n/a

Sequential read nearly doubles on Gen5; the B200 GPU complex is new to this node.

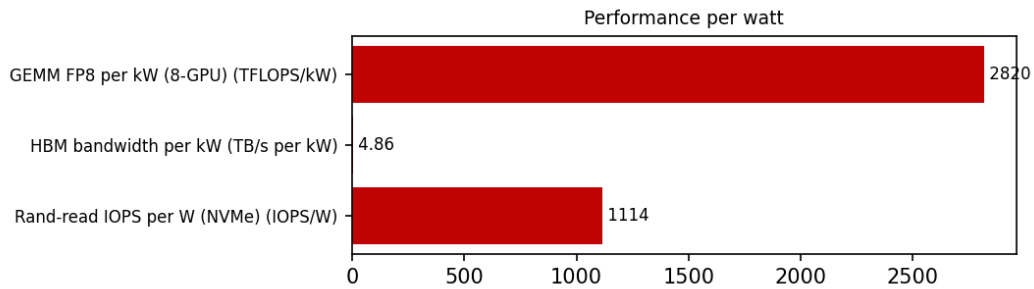
Power & Thermal Telemetry

Synchronized 1 Hz timeline across the run. Phase markers annotate idle baseline, load phases and cooldown. Peak wall power and inlet/outlet temperature occur during the combined-load phase, not in isolation.



Sampling achieved: 1 Hz (Redfish).

Efficiency -- Performance per Watt



Component Health

Component	Link	Temp pre->post	Wear	Media errors	Verdict
nvme0 PM9D3a	Gen5 x4	41 -> 58 C	0%	0	PASS
nvme1 PM9D3a	Gen5 x4	42 -> 59 C	0%	0	PASS
nvme2 PM9D3a	Gen5 x4	40 -> 57 C	1%	0	PASS
nvme3 PM9D3a	Gen5 x2 (!)	43 -> 61 C	0%	0	ACTION
8x B200 (HGX)	NVLink5	37 -> 71 C	n/a	0 ECC	PASS

BMC SEL (before → after): SEL cleared at run start; 0 new critical events after run (1 informational PSU input-history entry).

Kernel log (dmesg) diff: No new MCE, ECC, PCIe AER, NVMe reset or Xid GPU events. One pre-existing informational APST quirk message (unchanged).

Findings & Recommendations

Findings

- nvme3 negotiated a PCIe Gen5 x2 link (capable x4); 4K random write fell to 190k IOPS (53% of the 360k datasheet rating) and sequential write to 2.1 GB/s.
- All 8x B200 sustained ~4.41 PFLOPS FP8 (98% of datasheet) with NVLink at 1.76 TB/s and zero ECC errors; DCGM level-3 diagnostics passed on every GPU.
- Peak combined-load wall power was 12.6 kW at 25 C inlet / 52 C outlet with no GPU thermal throttling; the 5+1 PSU shelf retained redundancy throughout.

Recommendations

- Re-seat nvme3 and confirm slot-6 bifurcation is x4x4x4x4 in BIOS; if the x2 link persists, swap the riser and re-run the storage tier.
- Retain the current performance profile (C-states off, NPS2, ASPM off) for production AI workloads; it is the configuration these GPU/memory numbers were measured under.
- Monitor GPU3 (ran ~4 C warmer); within spec but worth tracking across the soak fleet.

Conclusion

Area	Outcome	Verdict
AI / GPU complex	8x B200 at 98% datasheet, full NVLink, 0 ECC, DCGM L3 pass	PASS
Acceptance criteria	Compute / memory / network meet or exceed datasheet; thermals in envelope	PASS
Storage	3 of 4 drives nominal; nvme3 link degraded -- remediate and re-test	ACTION

Area	Outcome	Verdict
Release recommendation	Approve for AI deployment after nvme3 PCIe remediation	LIMITED

Deployment & Rack Position

Physical placement of the system under test. Full rack-mount, rail-kit and mainboard service procedures are produced as a separate branded install guide.

Rack R12 (lab)

U41	
U39	
U37	
U35	
U33	
U31	
U29	
U27	ty-cam-38
U25	
U23	
U21	
U19	
U17	
U15	
U13	
U11	
U9	
U7	
U5	
U3	
U1	

Reproducibility Appendix

Exact tooling, kernel command line, BIOS state and a SHA-256 of the raw data bundle, so this run can be audited and reproduced.

Fio	fio-3.36
Nvme Cli	nvme version 2.8
Ipmitool	1.8.19
Stress Ng	0.17.06
Iperf3	3.16
Dcgmi	3.3.6
Cuda	12.6
Kernel	6.8.0-45-generic
Kernel Cmdline	ro quiet pcie_aspm=off nvme_core.default_ps_max_latency_us=0 iommu=pt nvidia_drm.modeset=1
Tuned Profile	accelerator-performance
Bios Profile	Max Performance (Determinism: Power)

Commands	see runs/run.log for exact invocations
Data Bundle Sha256	5418d6ae80f8f6f897e979caa0c8b759fd32b9d43a8e4b48cfae878be8d2ba29